

**ROLL NO:**

Roll No (Left Side):

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TU/CSE

Tezpur UniversitySessional test-1 (Spring 2023)**CO 218: Data Communication***Full marks=25**Time=45 min**(The figures in the right-hand margin indicate full marks for the questions. Question 1 to 10 carries 1 mark each)*

1. Consider a digitized voice channel is made by digitizing an analog voice signal offering bit rate of 64 kbps. If the system samples the signal at twice the highest frequency (two samples per hertz) having each sample using 8 bits Calculate the bandwidth of the analog voice signal? [3]
  
  
  
  
  
  
  
  
  
  
2. Assume that a link with implementing a low-pass channel offers bandwidth 100 kHz. What is the maximum bit rate of this channel and which harmonics will be in use to achieve the maximum bit rate? [2]
  
  
  
  
  
  
  
  
  
  
3. Consider the SNR of a channel is 24 dB. If the capacity of the channel is 8Kbps, calculate the bandwidth of the channel. [2]
  
  
  
  
  
  
  
  
  
  
4. What is the required bandwidth of a low-pass channel if we need to send at 2 Mbps of bit rate by using baseband transmission provided the 1<sup>st</sup>, 3<sup>rd</sup> and 5<sup>th</sup> harmonics are employed? [3]

5. Which statements are correct about circuit switch mechanism?
- For the duration of the conversation dedicated communication path is established.
  - A connected sequence of physical links form a path between nodes.
  - Incomming data are switched to outgoing channel with delay at each node.
  - A logical channel is assigned to the connection on each link.
- (A) i, ii, iv are correct      (B) i, iii, iv are correct      (C) i, ii, iii are correct      (D) ii, iv are correct      **ANS: \_\_\_\_\_**
6. Which of the following is not true for defining the term called protocol?
- A set of rules governing the format and meaning of the data units that are exchanged between peer entities within a layer.
  - Entities are free to change their protocols, provided they do not change the service visible to their user
  - Entities use protocols in order to implement their service definitions.
  - Protocols can be termed as set of service primitives which are operations of layers to above.
- ANS: \_\_\_\_\_**
7. Packet switching can be best described by the statement
- Data is split into multiple packets, each routed using multiple paths out of sequence and packets pass from node to node between source and destination.
  - Data is split into multiple packets, each routed using single path out of sequence and packets pass from node to node between source and destination.
  - Data is split into multiple packets, each routed using multiple paths in sequence and packets pass from node to node between source and destination.
  - Data is split into multiple packets, each routed using same path in sequence and packets pass from node to node between source and destination.
- ANS: \_\_\_\_\_**
8. Which statements are correct about transmission impairments?
- Signal received may differ from signal transmitted.
  - Signal quality suffers degradation in case of analog signal.
  - Possibility of bit error is always there in case of digital signal.
  - Possibility of bit error is always there in case of analog signal.
- (A) i, ii, iv are correct      (B) i, iii, iv are correct      (C) iii, iv are correct      (D) i, ii, iii are correct      **ANS: \_\_\_\_\_**
9. "Near the center frequency of a channel the velocity of the transmitted signal tends to be highest and fall off towards the two edges of the channel", is the property of
- Delay Distortion
  - Impulse
  - Intermodulation
  - Crosstalk
- ANS: \_\_\_\_\_**
10. Absolute bandwidth of a time-limited signal is
- Infinite and contains most the power
  - Infinite and effective band contains most the power
  - Finite and effective band contains most the power
  - All the above.
- ANS: \_\_\_\_\_**
11. Which of the below combinations are not correct to describe noise as impairment?
- Thermal noise and intermodulation.
  - Intermodulation and crosstalk.
  - Crosstalk and impulse.
  - Impulse and delay distortion.
- ANS: \_\_\_\_\_**
12. Which of the following is not a disadvantage of fragmentation?
- Fragmentation incurs shorter delay in data delivery
  - Fragmentation increases amount interrupt at the receiver
  - Fragmentation impose more overhead on fragment processing
  - Fragmentation increases processing time
- ANS: \_\_\_\_\_**
13. Which of the following is not a flow control scheme?
- Waiting window
  - Sliding window
  - Stop and wait
  - Credit System
- ANS: \_\_\_\_\_**
14. Optical fibre transmission implements \_\_\_\_\_ and uses \_\_\_\_\_ for efficient data rate.
- WDM and ILD
  - WDM and LED
  - FDM and ILD
  - FDM and LED
- ANS: \_\_\_\_\_**
15. Which statements are correct about the frequency band of a transmission system?
- The greater is the effective bandwidth of a frequency band, the higher is the data rate that is achievable
  - The higher the center frequency of a band, the higher is the potential bandwidth
  - The higher the center frequency of a band, the higher is the effective bandwidth of a band
  - The higher is the potential bandwidth of a band, higher the potential data rate
- (A) i, ii, iv are correct      (B) i, iii, iv are correct      (C) i, ii, iii are correct      (D) ii, iv are correct      **ANS: \_\_\_\_\_**
16. Addressing, Portability and reliable data exchange are the basic function of \_\_\_\_\_ and \_\_\_\_\_ layer.
- Data Link, Transport
  - Physical, Data Link
  - Transport, Application
  - Network, Transport
- ANS: \_\_\_\_\_**
17. Match the following.
- |                                  |                          |
|----------------------------------|--------------------------|
| i) LAN                           | a) Data link layer       |
| ii) Congestion Control           | b) broadcast system      |
| iii) Flow control                | c) Presentation layer    |
| iv) Client –Server               | d) Wait and stop service |
| v) Encoding Syntax and semantics | e) Network layer         |
| vi) Sequence Number              | f) Request reply         |
|                                  | g) Transport layer       |